

2D List Code Tracing

variables	values
matrix	$\begin{bmatrix} 1, 2, 3 \\ 4, 5, 6 \\ 7, 8, 9 \end{bmatrix}$

outside
for loop
iter: # 1

$$i = 0$$

$$j = 0 \Rightarrow \text{matrix}[0][0] = 1$$

$$\text{if } (0 + 0) \% 2 == 0:$$

$$1 * 2 = 2$$

*skip elif

$$\text{if } 0 + 0 \neq 3 - 1$$

variables

matrix

values =

$\begin{bmatrix} 2, 2, 3 \\ 4, 5, 6 \\ 7, 8, 9 \end{bmatrix}$

$$j = 1 \Rightarrow \text{matrix}[0][1] = 2$$

$$\text{if } (0 + 1) \% 2 \neq 0$$

$$\text{elif } 0 \neq 1$$

$$\text{if } 0 + 1 \neq 3 - 1$$

(no change to memory table)

$j=2 \Rightarrow \text{matrix}[0][2]=3$

if $(0+2) \% 2 == 0$:

$$3 * 2 = 6$$

* skip elif

if $0+2 == 3-1$:

$$6-1 = 5$$

var	val
matrix	$\begin{bmatrix} 2, 2, 5 \\ 4, 5, 6 \\ 7, 8, 9 \end{bmatrix}$

outside
for loop

iter. #2 $j=0 \Rightarrow \text{matrix}[1][0]=4$

if $(1+0) \% 2 \neq 0$:

elif $1 > 0$:

$$4 + 3 = 7$$

if $1+0 \neq 3-1$

var	val
matrix	$\begin{bmatrix} 2, 2, 5 \\ 7, 5, 6 \\ 7, 8, 9 \end{bmatrix}$

$j = 1 \Rightarrow \text{matrix}[1][1] = 5$

if $(1+1) \% 2 == 0$:

$5 * 2 = 10$

* skip elif

if $1+1 == 3-1$:

$10 - 1 = 9$

var	val
matrix	$\begin{bmatrix} 2, 2, 5 \\ 7, 9, 6 \\ 7, 8, 9 \end{bmatrix}$

$j = 2 \Rightarrow \text{matrix}[1][2] = 6$

if $(1+2) \% 2 \neq 0$

elif $1 \neq 2$

if $1+2 \neq 3-1$

outside
for loop
iter. #3

$i = 2$

$j = 0 \Rightarrow 7$

if $(2+0) \% 2 == 0$:

$7 * 2 = 14$

* skip elif

if $2+0 == 3-1$:

$14 - 1 = 13$

var	val
matrix	$\begin{bmatrix} 2, 2, 5 \\ 7, 9, 6 \\ 13, 8, 9 \end{bmatrix}$

$j = 1 \Rightarrow 8$

if $(2+1) \% 2 \neq 0$

elif $2 > 1$:

$8 + 3 = 11$

if $2+1 \neq 3-1$

var	val
matrix	$\begin{bmatrix} 2, 2, 5 \\ 7, 9, 6 \\ 13, 11, 9 \end{bmatrix}$

j = 2 ⇒ 9

if (2+2) % 2 == 0:

9 * 2 = 18

* skip elif

if 2+2 != 3-1

P, P, F

matrix[0][2] = 2 + 9 = 7

var	val
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matrix	[2, 2, 7], ...]
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matrix[2][1] = 13 * 2 = 26

var	val
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matrix	[..., [13, 26, 18]]
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matrix[1][1] = 2 + 18 = 20

var	val
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matrix	[..., [7, 20, 6], ...]
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print()

[2, 2, 7]

[7, 20, 6]

[13, 26, 18]